

Please check the examination details below before entering your candidate information

Candidate surname		Other names	
Pearson Edexcel Award		Centre Number	Candidate Number
Time 3 hours	Paper reference	9811/01	
Advanced Extension Award Mathematics			
You must have: Mathematical Formulae and Statistical Tables An insert for questions 5, 6 and 7			Total Marks

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- **Calculators may not be used.**
- You must **show all your working.**
- Answers should be given in as simple a form as possible, e.g. $\frac{2\pi}{3}$, $\sqrt{2}$, $3\sqrt{2}$.

Information

- The total mark for this paper is 100 of which **7** marks are for style and clarity of presentation.
- The style and clarity of presentation marks will be indicated as **(+S1) or (+S2)**.
- There are 7 questions in this question paper.
- The marks for each question are shown in brackets.
- The total mark for each question is shown at the end of the question.

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.
- Good luck with your examination.

Turn over ►

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1. $f(x) = x^{(x^2)} \quad x > 0$

Use logarithms to find the x coordinate of the stationary point of the curve with equation $y = f(x)$.

(5)



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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 5 marks)



2.

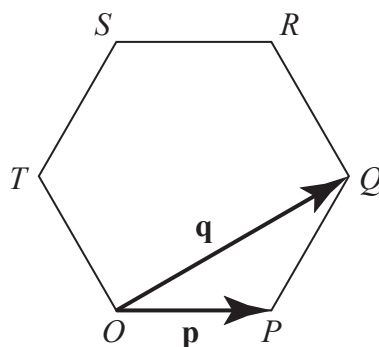


Figure 1

Figure 1 shows a regular hexagon $OPQRST$.

The vectors $\mathbf{p}$ and $\mathbf{q}$ are defined by $\mathbf{p} = \overrightarrow{OP}$ and $\mathbf{q} = \overrightarrow{OQ}$

Forces, in Newtons, $\mathbf{F}_P = (\overrightarrow{OP})$, $\mathbf{F}_Q = 2 \times (\overrightarrow{OQ})$, $\mathbf{F}_R = 3 \times (\overrightarrow{OR})$, $\mathbf{F}_S = 4 \times (\overrightarrow{OS})$ and $\mathbf{F}_T = 5 \times (\overrightarrow{OT})$ are applied to a particle.

- (a) Find, in terms of $\mathbf{p}$ and $\mathbf{q}$, the resultant force on the particle. (5)

The magnitude of the acceleration of the particle due to these forces is 13 m s^{-2}

Given that the mass of the particle is 3 kg ,

- (b) find $|\mathbf{p}|$ (5)



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Question 2 continued

Lined area for writing the answer to Question 2.



Question 2 continued

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Question 2 continued

Lined area for writing answers.

(Total for Question 2 is 10 marks)



3. (a) Use the formulae for $\sin(A \pm B)$ and $\cos(A \pm B)$ to prove that $\tan(90^\circ - \theta) \equiv \cot \theta$ (3)

- (b) Solve for $0 < \theta < 360^\circ$

$$2 - \sec^2(\theta + 11^\circ) = 2 \tan(\theta + 11^\circ) \tan(\theta - 34^\circ)$$

Give each answer as an integer in degrees.

(8)

(+S1)



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Question 3 continued

Handwriting practice area with horizontal lines.



Question 3 continued

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4. Given that $f(x) = e^{x^3-2x}$

(a) find $f'(x)$

(2)

The curves C_1 and C_2 are defined by the functions g and h respectively, where

$$g(x) = 8x^3e^{x^3-2x} \quad x \in \mathbb{R}, x > 0$$

$$h(x) = (3x^5 + 4x)e^{x^3-2x} \quad x \in \mathbb{R}, x > 0$$

(b) Find the x coordinates of the points of intersection of C_1 and C_2

(4)

Given that C_1 lies above C_2 between these points of intersection,

(c) find the area of the region bounded by the curves between these two points.

Give your answer in the form $A + Be^C$ where A , B , and C are exact real numbers to be found.

(7)

(+S1)



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Question 4 continued

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(Total for Question 4 is 14 marks)



5. An aeroplane leaves a runway and moves with a constant speed of V km/h due north along a straight path inclined at an angle $\arctan\left(\frac{3}{4}\right)$ to the horizontal.

A light aircraft is moving due north in a straight horizontal line in the same vertical plane as the aeroplane, at a height of 3 km above the runway.

The light aircraft is travelling with a constant speed of $2V$ km/h.

At the moment the aeroplane leaves the runway, the light aircraft is at a horizontal distance d km behind the aeroplane.

Both aircraft continue to move with the same trajectories due north.

- (a) Show that the distance, D km, between the two aircraft t hours after the aeroplane leaves the runway satisfies

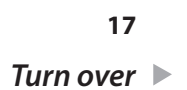
$$D^2 = \left(\frac{6}{5}Vt - d\right)^2 + \left(\frac{3}{5}Vt - 3\right)^2 \quad (3)$$

Given that the distance between the two aircraft is never less than 2 km,

- (b) find the range of possible values for d .



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Question 5 continued

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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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6.

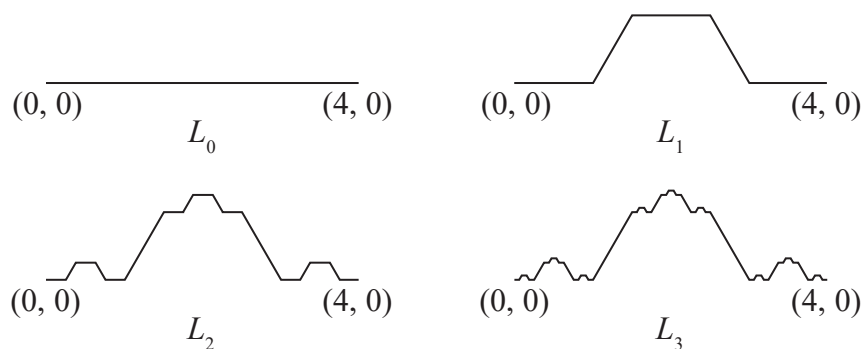


Figure 2

Figure 2 shows the first few iterations in the construction of a curve, L .

Starting with a straight line L_0 of length 4, the middle half of this line is replaced by three sides of a trapezium above L_0 as shown, such that the length of each of these sides is $\frac{1}{4}$ of the length of L_0 .

After the first iteration each line segment has length one.

In subsequent iterations, each line segment parallel to L_0 similarly has its middle half replaced by three sides of a trapezium above that line segment, with each side $\frac{1}{4}$ the length of that line segment.

Line segments in L_n are either parallel to L_0 or are sloped.

(a) Show that the length of L_2 is $\frac{23}{4}$ (2)

(b) Write down the number of

- (i) line segments in L_n that are parallel to L_0
- (ii) sloped line segments in L_2 that are not in L_1
- (iii) **new** sloped line segments that are created by the $(n + 1)$ th iteration. (3)

(c) Hence find the length of L_n as $n \rightarrow \infty$ (6)

The area enclosed between L_0 and L_n is A_n

(d) Find the value of A_1 (2)

(e) Find, in terms of n , an expression for $A_{n+1} - A_n$ (3)

(f) Hence find the value of A_n as $n \rightarrow \infty$ (3)



Question 6 continued

The same construction as described above is applied externally to the three sides of an equilateral triangle of side length a .

Given that the limit of the area of the resulting shape is $26\sqrt{3}$

(g) find the value of a .

(3)

(+S2)



Question 6 continued

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Question 6 continued

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Question 6 continued

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Question 6 continued

Handwriting practice area with horizontal lines.

(Total for Question 6 is 24 marks)



7. A circle C has centre $X(a, b)$ and radius r .

A line l has equation $y = mx + c$

(a) Show that the x coordinates of the points where C and l intersect satisfy

$$(m^2 + 1)x^2 - 2(a - m(c - b))x + a^2 + (c - b)^2 - r^2 = 0 \quad (2)$$

Given that l is a tangent to C ,

(b) show that

$$c = b - ma \pm r\sqrt{m^2 + 1} \quad (6)$$

The circle C_1 has equation

$$x^2 + y^2 - 16 = 0$$

and the circle C_2 has equation

$$x^2 + y^2 - 20x - 10y + 89 = 0$$

(c) Find the equations of any lines that are normal to both C_1 and C_2 , justifying your answer. (3)

(d) Find the equations of all lines that are a tangent to both C_1 and C_2

[You may find the following Pythagorean triple helpful in this part: $7^2 + 24^2 = 25^2$] (11)

(+S2)



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Question 7 continued

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Question 7 continued

Handwriting practice area with 30 horizontal lines.



Question 7 continued

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(Total for Question 7 is 24 marks)

**FOR STYLE AND CLARITY OF PRESENTATION: 7 MARKS
TOTAL FOR PAPER IS 100 MARKS**

